

Global Identifiability in a Three-Taxon LGT Model

Exhaustive genealogy calculation under JC69

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Principal theorem. Fix a known JC69 substitution rate $\mu > 0$. For one sampled ortholog from each taxon on the rooted clocklike species tree $((1, 2), 3)$, with $0 < t_1 < t_2$, constant ordered-pair transfer rate $\lambda > 0$, donor retention, recipient replacement, no incomplete lineage sorting, and exact population site-pattern probabilities, the exhaustive ordered-pair Poisson LGT map is globally injective. The distribution uniquely determines (t_1, t_2, λ) at every interior point, and the Jacobian has rank three everywhere. There is no exceptional interior set and no second interior preimage.

1. The omitted movement

A species branch is *ancestrally unoccupied* when it currently carries none of the ancestral lineages of the sampled leaf genes; the branch still carries a population gene copy.

Choose $0 < u < v < t_1$. At u , branch 1 transfers into branch 2. Backward in time, lineage 2 jumps to branch 1 and coalesces with lineage 1, fixing cherry $(1, 2)$. Branches 1 and 3 then carry the surviving sampled ancestry, and branch 2 is ancestrally unoccupied. At v , branch 2 transfers into branch 3. Backward in time, lineage 3 jumps to branch 2. This causes no immediate coalescence, but the surviving lineages now occupy branches 1 and 2 and coalesce vertically at t_1 . Without the second transfer they remain on branches 1 and 3 and do not coalesce at t_1 .

Thus the second transfer changes the second coalescence time and branch lengths without changing the already formed cherry. Topology frequencies can remain correct while site-pattern probabilities change.

2. Exhaustive CTMC correction

After the first sampled-gene coalescence, let the state be the identity of the ancestrally unoccupied branch. Conditional on no second transfer-coalescence, this state has generator

$$Q = \frac{\lambda}{2} \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}.$$

The second transfer-coalescence has hazard λ . Given survival for elapsed time s ,

$$P_{ee}(s) = \frac{1}{3} + \frac{2}{3}e^{-3\lambda s/2}, \quad P_{ee'}(s) = \frac{1}{3} - \frac{1}{3}e^{-3\lambda s/2}.$$

This sums arbitrarily many noncoalescing movements exactly. The resulting genealogy measure is disjoint, exhaustive, and normalized. It recovers

$$P(T_1) = \frac{1}{3} + \frac{2}{3}e^{-3\lambda t_1}, \quad P(T_2) = P(T_3) = \frac{1}{3} - \frac{1}{3}e^{-3\lambda t_1}.$$

3. Fourier and pattern coordinates

For a three-leaf ultrametric gene tree with interior times $g_1 < g_2$, the cherry-pair JC69 Fourier coordinate is $e^{-2\mu g_1}$,

either noncherry pair is $e^{-2\mu g_2}$, and the nontrivial three-way coordinate is $e^{-\mu(g_1+2g_2)}$.

For aggregate classes $p_0, p_{12}, p_{13} = p_{23}, p_D$, define

$$A = \frac{4(p_0 + p_{12}) - 1}{3}, \quad B = \frac{4(p_0 + p_{13}) - 1}{3}, \\ C = \frac{16p_0 - 1 - 3A - 6B}{6}.$$

This is an affine isomorphism, and

$$A - B = \frac{4}{3}(p_{12} - p_{13}).$$

4. Compact corrected map

Set

$$q = \frac{\lambda}{\lambda + \mu}, \quad x = e^{-(\lambda + \mu)t_1}, \quad y = e^{-(\lambda + \mu)(t_2 - t_1)},$$

then $0 < q, x, y < 1$. Put

$$r = q + (1 - q)y^2, \quad X = x^{2-q}, \quad Z = x^3,$$

and $D = A - B$, $M = (A + 2B)/3$. Exhaustive integration gives

$$D = (1 - r)x^{2+q/2}, \\ M = \frac{q(1 - X)}{2 - q} + \frac{1 + 2r}{3}X, \\ C = \frac{q^2((q - 2)Z + 3X - (q + 1))}{(q - 2)(q + 1)} \\ + rZ + \frac{q(1 + 2r)}{q + 1}(X - Z),$$

with $q < r < 1$. These formulas do not depend on the numerical value of fixed μ after reparameterization.

5. Global-univalence argument

Fix observed D, M . Eliminating r yields

$$M = \Phi_q(X) = \frac{q}{2-q} + \frac{2(1-q)}{2-q}X - \frac{2D}{3}X^{-3q/[2(2-q)]}.$$

For every $q \in (0, 1)$, $\Phi'_q(X) > 0$, $\Phi_q(X) \rightarrow -\infty$ as $X \downarrow 0$, and the observed $M < \Phi_q(1)$. Hence each q determines a unique formal X, x, r .

Let $h = (1-r)/(1-q)$. The formal branch has $r < 1$, hence $h > 0$, and physical feasibility is exactly $0 < h < 1$. Exact implicit differentiation shows $dh/dq > 0$ whenever $h \leq 1$. Every crossing of $h = 1$ is therefore upward; a later re-entry would require a downward crossing. Since the generating point is feasible, the feasible q -set is a nonempty interval.

The 2×2 minor is

$$\det \frac{\partial(D, M)}{\partial(x, r)} = x^{3-q/2}[2 - q(1+r)] > 0.$$

An exact factorization and convexity argument prove

$$\det \frac{\partial(D, M, C)}{\partial(q, x, r)} < 0$$

throughout the domain. Along the fixed- (D, M) branch,

$$\frac{dC}{dq} = \frac{\det \partial(D, M, C)/\partial(q, x, r)}{\det \partial(D, M)/\partial(x, r)} < 0.$$

Thus C selects a unique q , and then unique x, r, y and (t_1, t_2, λ) .

6. Topology and rate scale

The corrected map gives

$$A - B = (1-q)(1-y^2)x^{2+q/2} > 0,$$

so $p_{12} > p_{13} = p_{23}$. After relabeling, the uniquely largest matching-pair class identifies the rooted species-tree cherry; the theorem then identifies the parameters.

If μ is unknown, the exact invariance

$$(t_1, t_2, \lambda, \mu) \mapsto (ct_1, ct_2, \lambda/c, \mu/c)$$

leaves q, x, y unchanged. A time or rate scale must therefore be fixed. This does not weaken the theorem for fixed known μ .

7. Two distinct associated maps

Auxiliary map F_{table} . This is the fourteen-history map obtained by distinguishing the two one-transfer density types and treating the displayed transfer variables as absolute gene-tree times. A 384-bit outward-rounded Krawczyk certificate proves a second unique preimage of the exact target generated by $(q, x, y) = (1/2, 1/10, 3/5)$. Both Jacobians exclude zero. Therefore the image contains a nonempty open set of regular values with at least two preimages. No claim is made about almost every point of the entire image.

Distributed map F_{src} . For arXiv:2607.14653v1 and repository commit

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exact evaluation at $(q, x, y) = (1/2, 81/100, 1/10)$ gives positive normalized classes but

$$A_{\text{src}} - B_{\text{src}} = -\frac{57472173}{800000000}, \quad p_{xxy} - p_{xyx} = -\frac{172416519}{3200000000}.$$

Here $p_{12} = p_{xxy}$, $p_{13} = p_{xyx}$, and $p_{23} = p_{yxx}$ under the source's aggregate convention. Thus F_{src} reverses the process inequality. The source snapshot records immutable identifiers and hashes.

8. Verification

Run

`make verify-exact`

from the package root. It deterministically checks genealogy normalization, all symbolic integrations, the Fourier/site-pattern transformation, all 64 JC69 patterns, the global-univalence identities, topology and scale identities, the exact source diagnostic, and the MPFR Krawczyk certificate. Its final line is

ALL EXACT CHECKS PASSED

Monte Carlo is separated as `make audit-simulation`. The interval verifier reconstructs the target in $\mathbb{Q}(\sqrt{10})$ and does not reuse the discovery root search.